The Lifecycle and Clustering of LLM Benchmarks – A Data Driven Analysis

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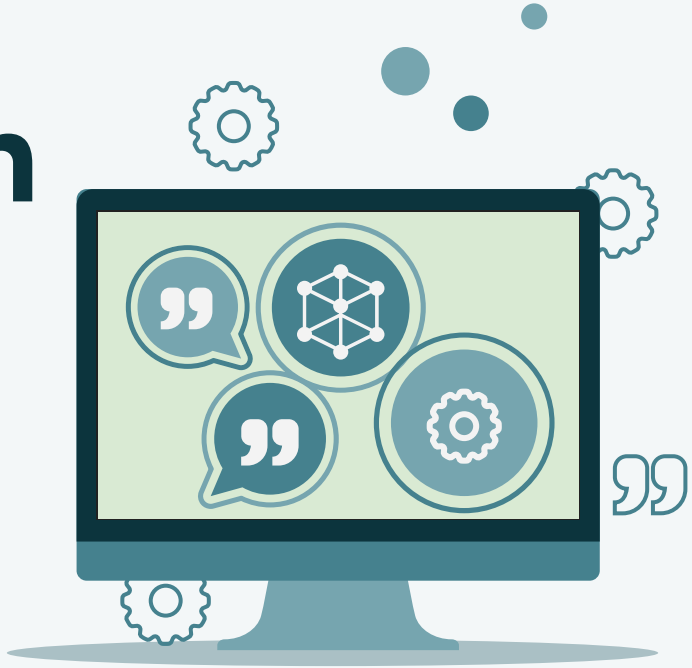
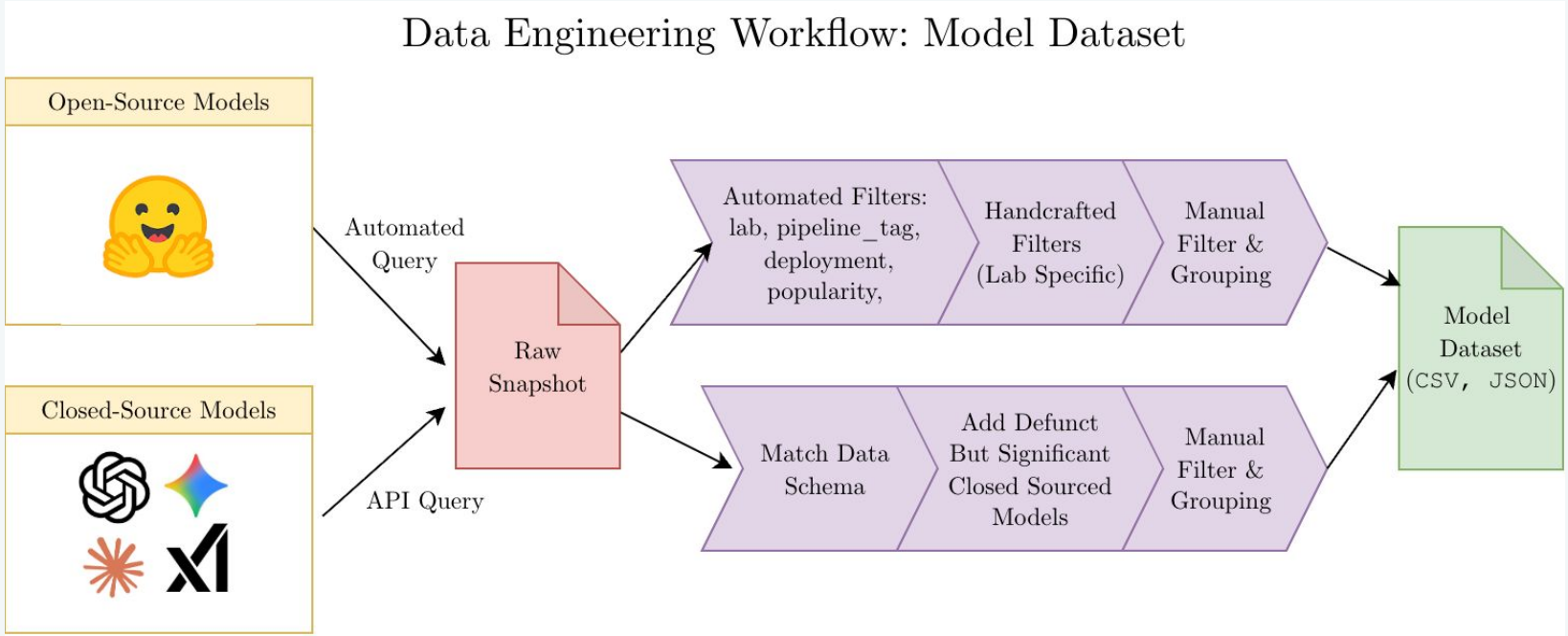
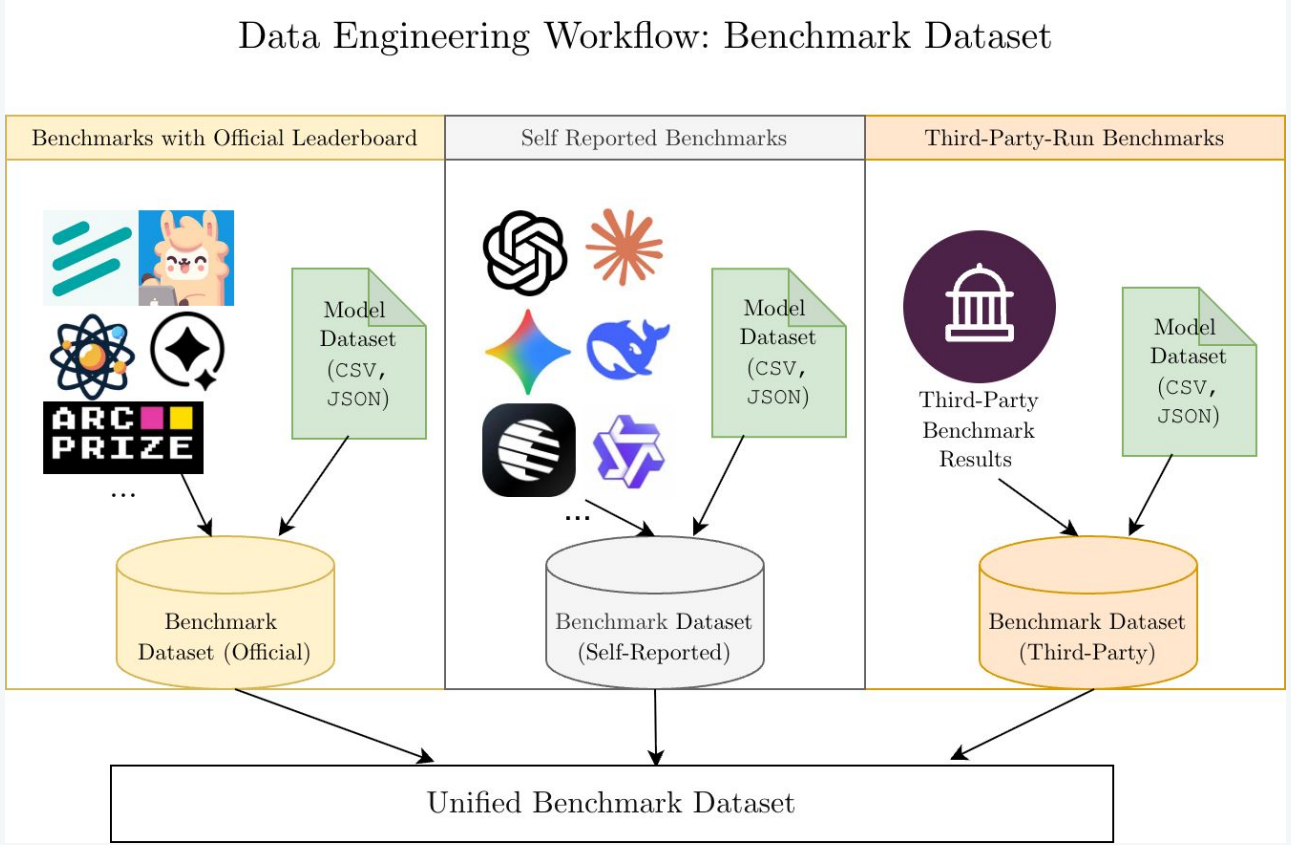
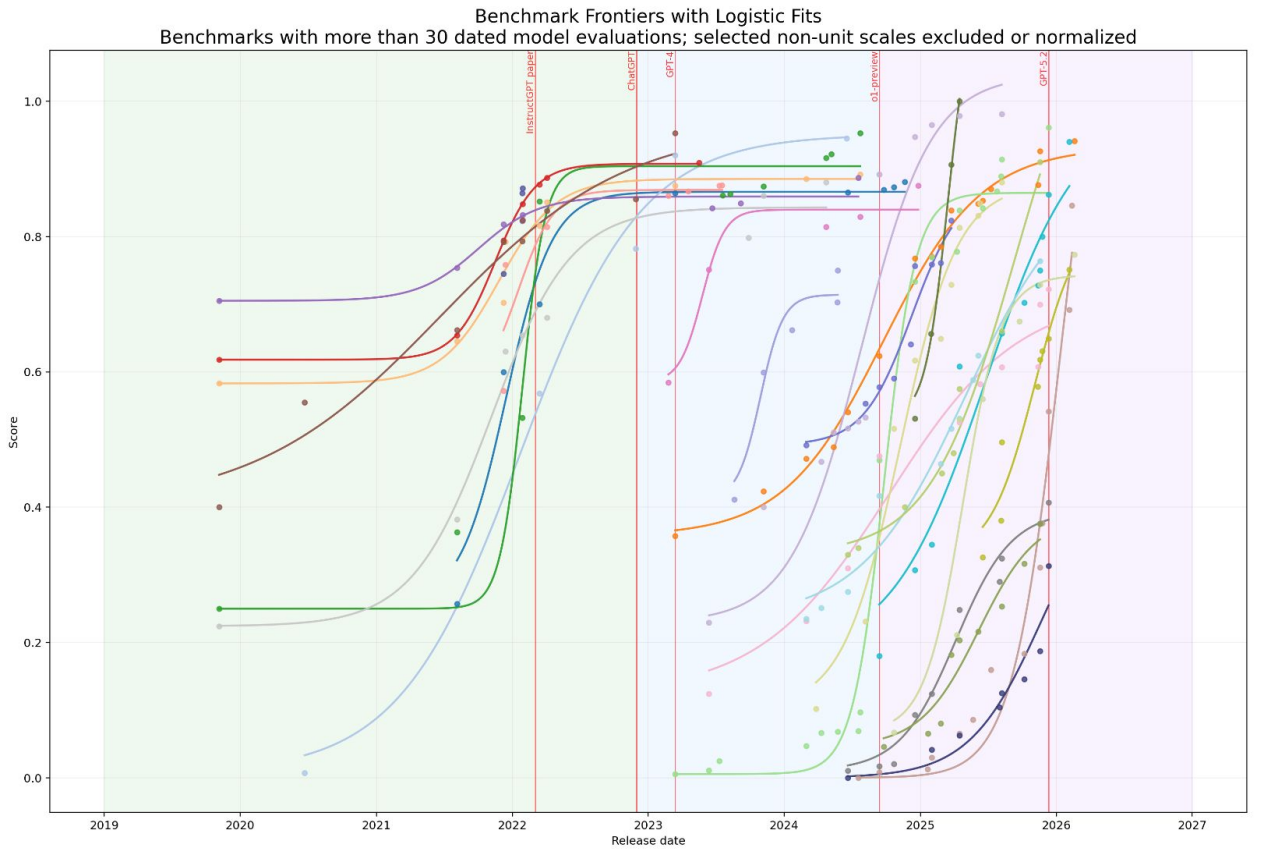
Data Preparation

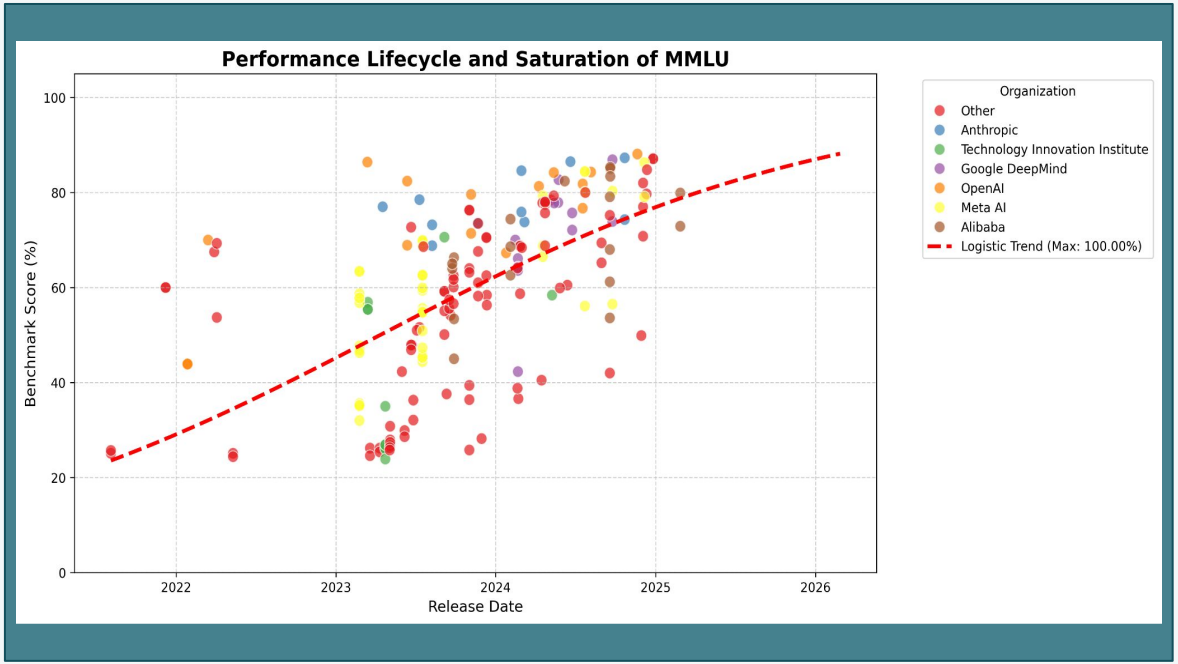



Data Engineering Workflow: Model Dataset
Open-Source Models
Closed-Source Models
Automated Query
API Query
Raw Snapshot
Automated Filters: lab, pipeline_tag, deployment, popularity,
Handcrafted Filters (Lab Specific)
Manual Filter & Grouping
Match Data Schema
Add Defunct But Significant Closed Sourced Models
Manual Filter & Grouping
Model Dataset (CSV, JSON)


Data Engineering Workflow: Benchmark Dataset
Benchmarks with Official Leaderboard
Self Reported Benchmarks
Third-Party-Run Benchmarks
ARC PRIZE
...
Model Dataset (CSV, JSON)
Benchmark Dataset (Official)
...
Model Dataset (CSV, JSON)
Benchmark Dataset (Self-Reported)
Third-Party Benchmark Results
Model Dataset (CSV, JSON)
Benchmark Dataset (Third-Party)
Unified Benchmark Dataset

Data Analysis

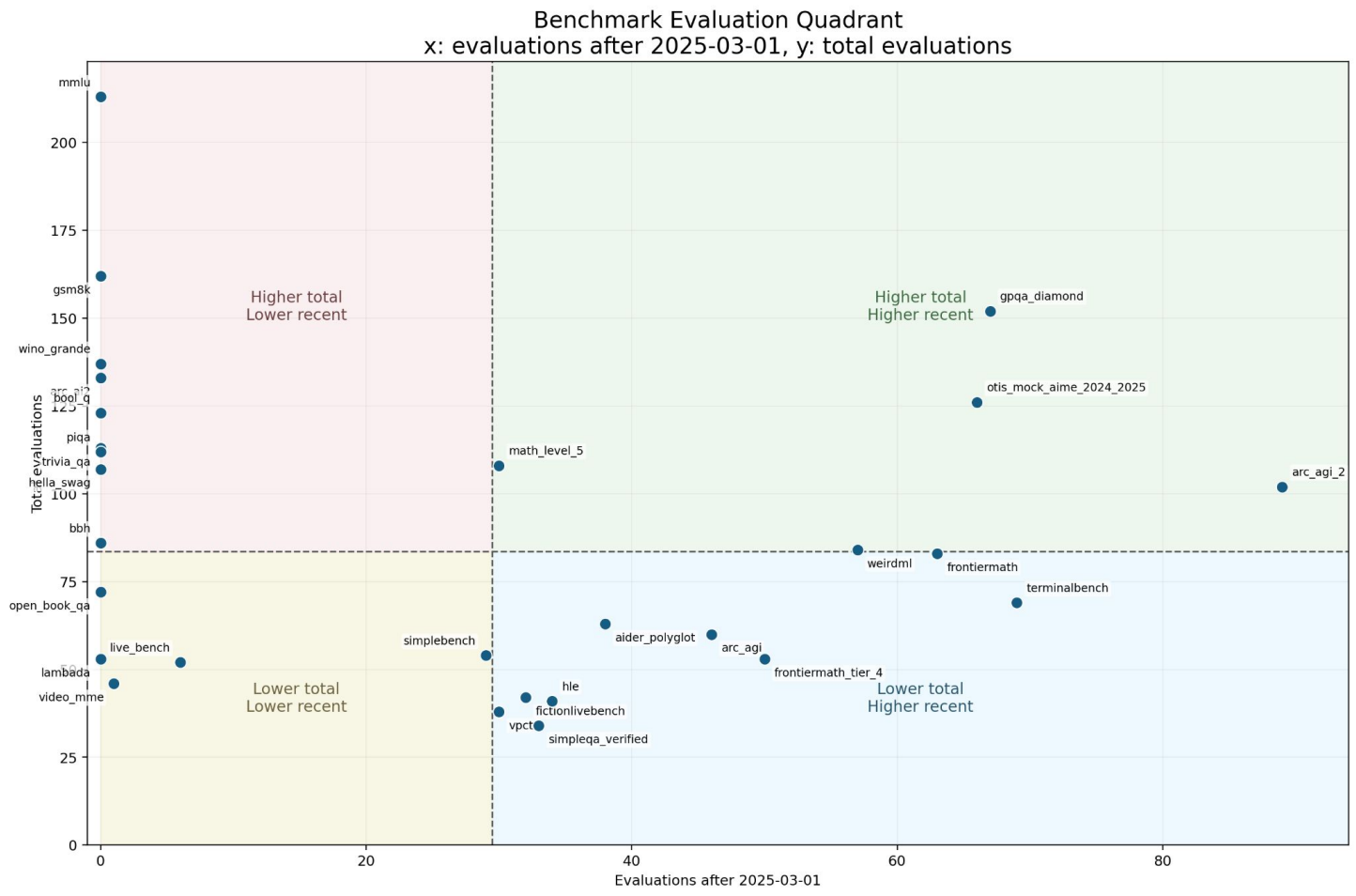

Benchmark Frontiers with Logistic Fits
Benchmarks with more than 30 dated model evaluations; selected non-unit scales excluded or normalized
Instruct GPT paper
ChatGPT
GPT-4
o1-preview
GPT-5.2
Score
1.0
0.8
0.6
0.4
0.2
0.0
2019
2020
2021
2022
2023
2024
2025
2026
2027
Release date

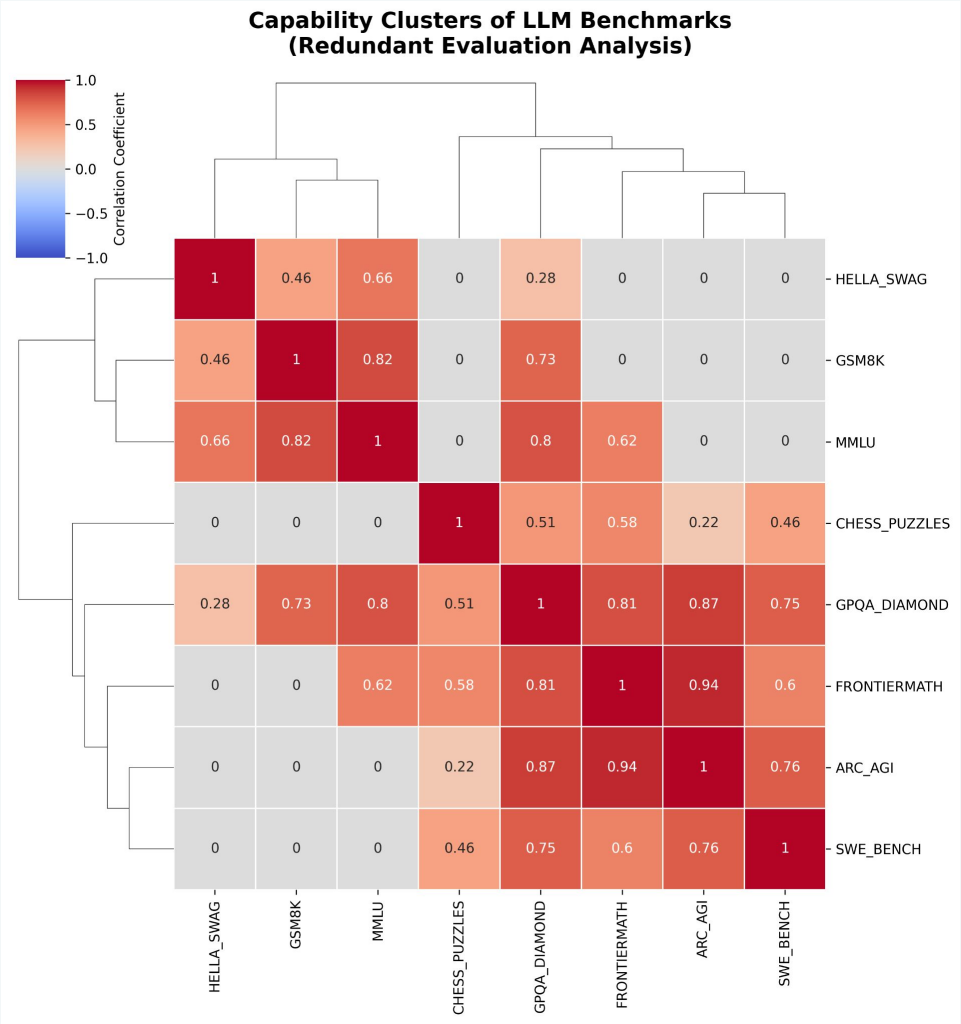


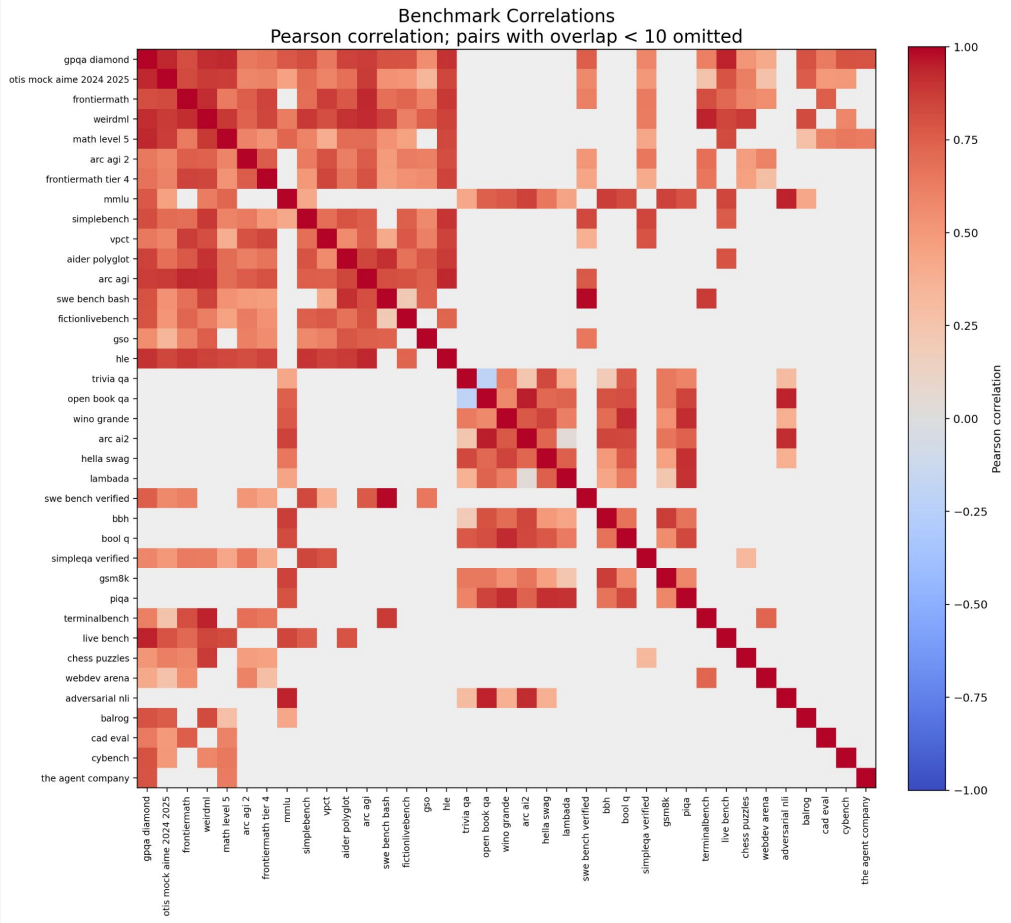
When benchmarks reach saturation:

1. New models show minimal performance differences

2. The benchmark loses discriminative power


Benchmark Evaluation Quadrant
x: evaluations after 2025-03-01, y: total evaluations
Higher total
Lower recent
Higher total
Higher recent
Lower total
Lower recent
Lower total
Higher recent
mmlu
gsm8k
wino_grande
bool_q
piqa
trivia_qa
hella_swag
bbh
open_book_qa
live_bench
lambada
video_mme
gpqa_diamond
otis_mock_aime_2024_2025
math_level_5
arc_agi_2
weirdml
frontiermath
terminalbench
simplebench
aider_polyglot
arc_agi
frontiermath_tier_4
hle
fictionlivebench
vpct
simpleqa_verified
Total evaluations
Evaluations after 2025-03-01
0
25
75
100
125
150
175
200
0
20
40
60
80


Capability Clusters of LLM Benchmarks
(Redundant Evaluation Analysis)
Correlation Coefficient
1.0
0.5
0.0
−0.5
−1.0
HELLA_SWAG
GSM8K
MMLU
CHESS_PUZZLES
GPQA_DIAMOND
FRONTIERMATH
ARC_AGI
SWE_BENCH
1 0.46 0.66 0 0.28 0 0 0
0.46 1 0.82 0 0.73 0 0 0
0.66 0.82 1 0 0.8 0.62 0 0
0 0 0 1 0.51 0.58 0.22 0.46
0.28 0.73 0.8 0.51 1 0.81 0.87 0.75
0 0 0.62 0.58 0.81 1 0.94 0.6
0 0 0 0.22 0.87 0.94 1 0.76
0 0 0 0.46 0.75 0.6 0.76 1


Benchmark Correlations
Pearson correlation; pairs with overlap < 10 omitted
gpqa diamond
otis mock aime 2024 2025
frontiermath
weirdml
math level 5
arc agi 2
frontiermath tier 4
mmlu
simplebench
vpct
aider polyglot
arc agi
swe bench bash
fictionlivebench
gso
hle
trivia qa
open book qa
wino grande
arc ai2
hella swag
lambada
swe bench verified
bbh
bool q
simpleqa verified
gsm8k
piqa
terminalbench
live bench
chess puzzles
webdev arena
adversarial nli
balrog
cad eval
cybench
the agent company
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
Pearson correlation

Thanks!